

NOVEMBER 01 1950

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J. Acoust. Soc. Am. 22, 701–707 (1950)

<https://doi.org/10.1121/1.1906674>





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frequency distribution of vowels in the monosyllabic words. We have investigated several languages from this point of view. Figure 6 shows three examples.

For German, we have divided the phonemes $|e|$ and $|o|$ into their two qualities $[e, \epsilon]$ and $[o, \circ]$ respectively. This was not done for the front rounded vowels where the different qualities are phonemic in the pronunciation of some native speakers. In a later version of our investigation we shall take these factors into consideration. We were mainly interested in the comparison of qualities of the vowels from the point of view of their production. We have also omitted the analysis into distinctive features which might make possible a more expressive mode of formulation.⁴

⁴ See paper by Lotz, "Speech and Language," elsewhere in this volume. I express my appreciation to Professor Lotz also for details communicated to me directly.

In French, the palatal vowels, namely $[a, e, i]$, dominate. In English, the vowels are crowded in the neighborhood of $[a]$ and these vowels are relatively similar to each other. In German, on the other hand, the distribution is very different. There is a large difference between the predominant vowels. The dominating vowels in German are $[a, i, u]$, that is to say, the most remote points of the vowel-triangle. This agrees well with the above-mentioned precision, characteristic of Germans.⁵

Probably this vowel arrangement of the three languages depends simply on the basis of articulation.

Other investigations along these lines have been begun, but not yet finished.

⁵ The overlapping of the circles depends only on the value of the relative frequencies; it has no phonetic value.

Description of Language Design

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(Received August 8, 1950)

Physicists describe speech with continuous mathematics, such as Fourier analysis or the autocorrelation function. Linguists describe language instead, using a discontinuous or discrete mathematics called "linguistics." The nature of this odd calculus is outlined and justified here. It treats speech communication as having a telegraphic structure. (Non-linguists normally fail to orient themselves in this field because they treat speech as analogous to telephony.) The telegraph-code structure of language is examined from top to bottom, and at each of its several levels of complexity (compared to the two levels of Morse code) its structure is shown to be defined by possibilities and impossibilities of combination among the units of that level. Above the highest level we find, instead of such absolute restrictions, conditional probabilities of occurrence: this is the semantic field, outside linguistics, where sociologists can work. Below the lowest level we find, instead of such absolute restrictions, conditional probabilities of phonetic quality: this is the phonetic field, outside linguistics, where physicists can work. Thus linguistics is peculiar among mathematical systems in that it abuts upon reality in two places instead of one. This statement is equivalent to defining a language as a symbolic system; that is, as a code.

THE word "design" may be interpreted in at least two senses without violating engineering usage. First, there is the work of the engineer who is designing something, for example a telephone system. Second, there is the finished something, such as that same telephone system, considered now as something to be analyzed and described. Our description of it, then, can also be called its "design." It is in this last sense that the word "design" has to be understood when we are talking about the design of language and how it is analyzed for description.

Engineers never have occasion to analyze a gadget the way linguists do a language. The nearest thing to it might be this sort of episode: A supplier had been asked to furnish a band-pass filter, single T section, with rather modest cut-off requirements. Instead of the expected classical constant- k filter—two series arms, one parallel arm, total three coils and three condensers

—he furnished a filter with three coils and four condensers connected in an unexpected way. The writer was asked to analyze the design and report whether it was worth adopting for future routine construction of band-pass filters. It turned out that the odd design had two advantages: one of the coils could be a large stock-size coil used without trimming, and the design gave any desired impedance-transformation without a transformer. But—and here is the crucial point—it did not seem to be possible to arrive at these conclusions without taking into consideration the designer's intentions; and the experience seemed to show that a "why" question has an answer only when intention is in the picture—when the presence of intention is given, so to speak, as a boundary-condition.

Linguists have described numerous languages, and have observed language design-changes through the centuries. But although quite a body of "how" knowl-

edge concerning these matters has been built up, we have so far nothing but a number of unconfirmable guesses concerning the "why." Hence most of us are now pretty sure about one thing: there is nothing conscious, nothing deliberate, about language-design. (Naturally we don't mean to study Basic English or Esperanto until we have first built up the theory of natural language design.) Accordingly we do not present the design of any language as the result of intentions—we do not answer "why" questions about the design of a language.

Having thus limited our problem, we proceed with greater confidence to look for answers to the "how" questions of language-design. In other words, we try to describe precisely; we do not try to explain. Anything in our description that sounds like explanation is simple loose talk—deliberately loose, perhaps, for the sake of persuasion by analogy—and is not to be considered part of current linguistic theory.

We can allow other people—telephone engineers or sociologists, for example—to speak artistically, imprecisely, about language. But as linguists we lay upon ourselves the condition that we must speak precisely about language or not at all. We can do that, of course, only under two conditions, of which the first has already been hinted at. First, we must limit our field, leaving outside it certain things to be treated precisely by engineers or by sociologists, while we speak of them more or less artistically. Second, within our field we must adopt a technique of precise treatment, which is by definition a mathematics. We must make our "linguistics" a kind of mathematics, within which inconsistency is by definition impossible. True, miscalculations may occur through inadvertence. And different workers describing the same language may arrive at conflicting statements because they have started out from different sets of axioms. But in principle every possible statement of ours must be either true or false—nothing half-way.

Have linguists succeeded in setting up such a mathematical style for describing language designs? Well, not quite; but our science is still young. In its mathematical phase it is just a quarter of a century old, for we date it from Bloomfield's "Set of Postulates for the Science of Language" (in the journal *Language* in 1925). And even physics has not yet entirely resolved the conflict between quantum theory and the wave theory of light. But of all the sciences and near-sciences which deal with human behavior, linguistics is the only one which is in a fair way to becoming completely mathematical, and other social scientists are already beginning to imitate the strict methods of the linguists.

Presenting only "how" statements without regard to "why," we of course have no assurance that we are "right" in the sense that would satisfy a theologian. And we feel that our descriptive statements fit actual speech-behavior, but we have no right to claim that they are "correct" in the sense that they fit the neural

events in the brains of the speaker and listener. Such arrogant modesty is not peculiar to linguists; it is simply the normal attitude of mathematicians confronting the real world. The mathematical description for reality has been most illuminatingly called a "map." Now when one holds a map (in the ordinary sense) in one's hands, one may say, "I feel that this is a map of the countryside around me here," but there is no way to prove logically that it is not instead a map of a piece of Australia or perhaps of some imaginary Treasure Island. Undeterred by the impossibility of logical justification, explorers use maps, scientists use mathematics, and linguists use the descriptive technique called "linguistics." All three have the same attitude toward the map. One proceeds across the terrain and simultaneously traces a line across the map; one notes discrepancies between the map and the direct apprehension of the real world, until the discrepancies seem to form a pattern themselves; then one corrects the map and starts all over again. All this is intuitive behavior, and logically unjustifiable. Nor does it need justification. The place for logic is inside the map, not between the map and the real world.

Ordinary mathematical techniques fall mostly into two classes, the continuous (e.g. the infinitesimal calculus) and the discrete or discontinuous (e.g. finite group theory). Now it will turn out that the mathematics called "linguistics" belongs in the second class. It does not even make any compromise with continuity, as does statistics, or infinite-group theory. Linguistics is a quantum mechanics in the most extreme sense. All continuity, all possibilities of infinitesimal gradation, are shoved outside of linguistics in one direction or the other. There are in fact two such directions in which we can and resolutely do expel continuity: semantics and phonetics.

Every language has "meaningful" molecules called "morphemes," sub-assemblies which roughly correspond to subdivisions of the real world outside language. The word *nose*, for example, consists of one such morpheme; the word *noses* consists of two, the second having as its real-world correspondent the conventional category of numerosness, one of our customary subdivisions or categorizations of the real world. Now consider these facts. The English word *nose* may refer to a part (how much?) of an airplane. With a lipstick draw a loop on your own face enclosing your nose and nothing else; another person will say that you have enclosed either too much or too little. Say in English: "The councillors all put their glasses on their noses" and then get the sentence translated into German; for English *noses* you will get the singular noun-form *Nase*, not the plural *Nasen*. The German knows that numerous councillors have equally numerous noses, and he has a word for "noses," but in this sentence he uses his word for "nose" instead. In linguistic terminology we simply say that the form *Nase* belongs to the category which we have called "singular." We chose that name the way

the physicist chose the name "work," because in German the category most often refers to lack of numerosness in the things referred to; but this is not a logical reason, it is only a motivation out of convenience. High-school physics teachers often have good reason to regret the choice of the term "work" for force times distance, and in expounding linguistic theory we often have the same reason for regretting the choice of terms "singular" and "plural." The trained listener, the listening linguist in this case, is not deceived; he knows that in calling *Nase* "singular" we have not said whether one nose or more than one nose was involved.

We keep on and say a few thousand things like this, but as long as we are talking about German or English we never place a noun-form in a third category side by side with "singular" and "plural," say a category which is neither singular nor plural, or is both, or is ambiguous as to number. The listening linguist, noting the absence of this other kind of statement, sooner or later decides that in German or in English every noun-form is either singular or else plural—*tertium non datur*. This may be called the tyranny of linguistic categories. It is found in every language. It is not a question of "correctness" in the popular sense of the word, for every kind of speech, elegant or sub-standard, has its own categorical tyrannies. For example, in the writer's dialect the form *whom* cannot follow a pause: it is permissible to say "To whom did you give it?" or "Who did you give it to?" but not "Whom did you give it to?" In situations where the normal language-design is not quite adequate, such as in the composition of a paper like this, we get amusing difficulties, such as finding it necessary to say either "one nose or more than one nose was involved" or else "one nose or several noses were involved." The difficulty is just as severe if we say "one sheep or more-than-one/several sheep was/were involved." That is, it is the category that tyrannizes over us, not the form, and not the meaning either; and even a word like *sheep* must be, each time that it is used, either singular or else plural. We might like to replace *was/were* with a verb-form which was, like the Irishman's trousers, singular at the top and plural at the bottom, but we haven't got any such category in English. In fact, in every known language there is a limit to the kinds and numbers of its categories, a limit which may seem very low compared to the elaborations which philosophers and poets indulge in. Then when an unusual message is to be transmitted, the categories are never split or supplemented, but are instead used in unusual combinations. The listener's total reaction may or may not be as expected, but his detail reactions are predictable at least to this extent: For every form-class represented in the utterance ("noun" is one form-class, "verb" is another) there are a certain number of dimensions of categorization (in the English noun there are two: possessive-*versus*-common, and singular-*versus*-plural); and now, for every dimension thus introduced into the

utterance, the listener will react to exactly one of the categories in that dimension—he will not react to more than one, or to none, whether the utterance is ambiguous or not. The reader can check this fact easily. Without looking back to the sentence about the councillors, let him ask himself whether it referred to present time or to past time. He will find that he has made up his mind one way or the other; and yet, on checking the wording, he will find that he was not told. The English categorization, past-*versus*-present, forced him to decide, even though the decision had to be made at random. Each language does this within the range of its own categories, *which will not be the same range as in our language*. In Chinese, for example, the translation lacks both the past-*versus*-present dimension and the singular-*versus*-plural dimension; these "meanings" can be indicated on occasion, but when they are not, the Chinese listener leaves the question open—he says he is not even aware that it is a question.

The linguistic categories, then, are absolutes which admit of no compromise. They correspond roughly to favorite categorizations in the real world, and it is generally held that every community subdivides the phenomena in the real world according to the categories of its language, rather than the reverse. But the correspondence between the discrete categories of the language and the continuous phenomena of the real world is not and cannot be precise. Our reaction, as linguists, to this situation, is very simple: all phenomena, whether popularly regarded as linguistic (such as the tone of anger in an utterance) or not, which we find we cannot describe precisely with a finite number of absolute categories, we classify as non-linguistic elements of the real world and expel them from linguistic science. Let sociologists and others do what they like with such things—we may wish them luck in their efforts to describe them precisely in their own terminology, but no matter whether they describe them with discrete categories or not, for us they remain vague, protean, fluctuating phenomena—in a word, for us they represent that "continuity" which we refuse to tolerate in our own science.

Have we ever had any choice? Could we have chosen to use a continuous rather than a discrete mathematics for describing language-design? Or perhaps an ambiguous mathematics which can be read both ways? Our experience with the analysis of a few hundred of the world's several thousand languages makes us very sure that we must answer negatively. When we hear of an imaginative child who has identified the "too" of "this one is too big" with the "two" of "two big ones" and then has experimented with saying "one-and-a-half big" and "three big" by way of weakening or strengthening the notion of "too big," we recognize that this is not the way languages work and we decide that that child was not only imaginative but probably also imaginary.

So much for the semantic end of language, with its

intuitive bridge between language and the real outside world, the bridge across which we banish continuity. We do exactly the same thing at the other end, the phonetic end of language. But there we are not dealing with philosophers; in the phonetic field we have to put our case to telephone engineers, experimental phoneticians, and other hard-headed citizens who will not be easy to convince. Let us state our view first, then discuss it.

Here we have "phonemics" inside linguistic science; and "phonetics," which is not so very different from and may even be considered a division of the physicist's "acoustics," in the realm of overt speech-behavior, which is not the realm of linguistics but belongs to the real world outside linguistics. When a physicist analyzes speech he finds a continuum of articulatory activity, of sound, and of events in the ear—at least until, working both ways from the sound, he comes to the all-or-none discharging of single nerve-fibers—and for describing any or all of this he uses a continuous mathematics, for instance Fourier analysis or the autocorrelation function. On the other hand, linguists have found it appropriate always to describe every language so far studied as having, and every act of speech as corresponding to, combinations of an astonishingly small number of smallest signalling units called "phonemes" which recur identically insofar as they can be said to occur at all. Thus the word *hotel* cannot be spoken the same twice, from the physicist's viewpoint, no matter whether the same person or different persons should utter it; and yet from the linguist's viewpoint it always has in the middle a phoneme /*t*/ which is identically the same every time. And it is not considered identical through neglect of unimportant variations in reality; rather, it is identical by definition, simply because it is a linguistic atom, a category, and these are either identically the same or absolutely different.

Here, then, we have another case where linguistics insists upon being atomistic and banishes continuity to the world outside language. Is this the right and only thing for a linguist to do? Telephone engineers have been known to protest at this point and say something like this: "When you were discussing linguistic discreteness and non-linguistic continuity in the semantic field you had me where you wanted me, for how should an engineer know anything about semantics? But now you are talking about sound, and here I am perfectly at home; I have even made extensive and precise observations upon normal speakers and listeners: for instance, I have gathered statistics about the variations in vowel qualities. Your mathematically identical phonemes are a delusion; what you've got are statistical norms, clusterings around averages which you take as norms and then arbitrarily make into absolutes. We are both talking about phonetics; and when you admit that I can prove continuity, you can't claim to prove discreteness.

"Let us agree to neglect the least important features

of speech sound, so that at any moment we can describe it sufficiently well with n measurements, a point in n -dimensional continuous space, n being not only finite but a fairly small number, say six. You agree? Good. Now the quality of the sound becomes a point which moves continuously in this 6-space, sometimes faster and sometimes slower, so that it spends more or less time in different regions, or visits a certain region more or less often. In the long run, then, we get a probability-density for the presence of the moving point anywhere in the 6-space. This probability-density varies continuously all over the space. Now wherever you can find a local maximum of probability-density, a place where the probability-density does not increase for short distances in any direction away from the place in question, I will let you call that place a 'phoneme'; and I expect, just as you do, that there will be not only a finite but a fairly small number of such points, say less than a hundred. Thus you see that there is no need to use a discrete mathematics to get a finite number of phonemes out of the phonetic continuum."

We could answer that our name for that would be, not "phoneme," but "allophone," and allophones belong to phonetics while phonemes belong to linguistics. But that would be evading the issue. The engineer's argument is a formidable one, and deserves to be met squarely. According to our view concerning the relationship between any mathematical map and the real phenomena to which it corresponds, it is not possible to argue logically in this field; therefore we have to argue by means of analogy. Fortunately a suitable analogy lies ready to hand.

Let us consider telegraphy, using Morse or Baudot coding. The units are the mark and the space; it doesn't matter whether the "space" is an electrical impulse of opposite polarity to the "mark," or an electrically neutral period. A Morse dash could be considered as three integrally sequent marks, and a letter-space as three spaces; then the units differ in only one dimension. Or the dash and the letter-space may be considered as independent signals; then there are four signals, differing from each other in two dimensions, namely electrical intensity (or polarity) and duration. Thus there are at least two ways of describing the Morse code, both discrete and both correct, each by its own axioms.

A Morse *S* is mark-space-mark-space-mark. A Baudot *Y* is mark-space-mark-space-mark—shall we say "also"? That depends; we can settle it later. At the far end of a long line, the marks and spaces are slurred together into a continuous smooth fluctuation so that on first inspection it may be far from easy to tell what the original signals must have been. Then a regenerative repeater is used to square up the marks and spaces so that they can be read or will operate an ordinary teletypewriter. But if the line is very long, not much noise will be needed to cause occasional errors in the re-

peater's operation. A regenerative repeater for Baudot signals will in general not handle Morse signals, or vice versa.

Now let us find a telephone engineer who knows nothing about telegraphy, and give him the output of a long telegraph line. What does he do with it? He does Fourier analysis and autocorrelation and perhaps something else—at any rate, at first he works on it with continuous mathematics, for the material is obviously continuous. But as long as he does that, he can make no sense out of it. In desperation he next differentiates the output voltage, so that he has two data for every instant, the voltage and its first time-derivative. Now for each five milliseconds he averages each of these, and plots the two averages as a single point on a sheet of graph-paper. When he has put a few thousand points on his paper, he sees a continuous variation of point-density, with a number of local maxima—perhaps a few dozen of them. To each maximum he assigns a label, a letter or a number, arbitrarily. Now he interprets his original data, his voltage and derivative data, in terms of these labels: every time the line-output and its derivative come close to one of those maximum-density points, he will write down the label of that point, so that finally he has replaced the original data by a string of those labels. Now he can turn this result over to a cryptanalyst, who will be able to read the original plain-language text out of it—not quite completely on first reading, but probably he will be able to reconstruct a good text if he knows the language well. Ridiculous? Not at all. This is exactly what our telephone engineer wanted to do with phonetics and phonemics.

Now we also know what that same telephone engineer will do after we tell him of the existence of telegraph codes employing molecular signals made up of atomic units, each unit being identically one or another of a very small number of atoms. He will not use Fourier analysis or any other continuous mathematics at first. Rather, he will start out by looking for wave-trains that look nearly alike, their slight differences being explicable, as he can easily see, either as the effects of noise or as the effects of context. He will see many occurrences of a three-peaked wave-train, no two occurrences precisely alike. But when the first peak is rather low, then there will be a long hollow before it; when there is no hollow, the first peak is as high as the second or a bit higher. This is a context-effect. Once the context-effects have been spotted, the remaining variations will be ascribed to noise. In this fashion he will learn to identify the three-peaked wave for Morse *S* or Baudot *V*, and also all the other common molecular signals.

Next he will set up the hypothesis that there are just n atomic signalling units, and that not only the inter-molecular context-effects but also the intra-molecular smoothings can be accounted for by low-pass filtering. Here he can use continuous mathematics for a while, and the result will be that this hypothesis

works best if n equals two. Now it won't take him long to reconstruct the entire telegraphic code; quite easily if it is Baudot, not so easily or swiftly if it is Morse, but in either case with absolute certainty.

Finally he will describe the design of the telegraphic code, and he will describe it in terms of absolutely identical square-wave signalling atoms, even though he has never observed anything that could be called absolute identity, and even though he knows that square waves are impossible. And when he has thus finished his job, he will be able to convince his colleagues, the other telephone engineers who knew nothing about telegraphy, that this is the right way to do it. Convince them by logic? Not a chance. Nothing but the superior elegance of his results will speak for the rightness of his method, but that will be enough.

Linguists find themselves so successful in describing languages in this fashion that they have elevated their descriptive technique to the rank of a theory about the nature of language. They say, in effect, that the design of any language is essentially telegraphic—that the language has the structure of a telegraph-code, using molecular signals made up of invariant atoms, and differing from e.g. the Morse code principally in two ways, one of which has already been hinted at: the codes called "languages" have numerous layers of complexity instead of only two, and in each layer there are severe limitations upon the combinations permitted. This much, we find, all languages have in common. But the particular code is different from one language or dialect to another, as the Baudot and Morse codes differ. Those two telegraphic codes share, on the phenomenal level, this molecular signal among others: mark-space-mark-space-mark. Now suppose we find this signal in the output of one telegraph line, and find it again in the output of another telegraph line; are they the same or are they not the same? That depends. If they are both Morse or both Baudot, then they are identically the same signal, no matter whether or not they are distorted by different noises or different line characteristics, or are sent at different speeds, or by a well-adjusted or a badly-adjusted transmitter. If one is Morse and the other is Baudot, then they are not in the least the same signal, even when their oscillograms come out the same within the limits of instrumental error.

This is obvious to any telegraph engineer. But in the linguistic field it did not seem obvious or even credible to a certain engineer who said to the writer: "Surely you can tell when two men pronounce the word *father* alike!" The answer had to be: "On the data you have given me, the question has no meaning. First I have to find out whether they rhyme *father* with *bother*. If they both do, or neither does, then I must find out whether both of them or neither of them pronounces *cot*-like *caught*. Then I have a few more questions of the same sort, with which I explore their contrasts for some distance around the word *father* in all directions. If I

always get the same answer from both of them, then I say, 'Yes, they pronounce *father* alike,' meaning 'identically the same,' even if they sound quite different when they say the word. But if, during this exploration, I find a discrepancy between their systems of phonemic contrasts before I have gone very far from the word *father* itself, then I shall have to say, 'No, they pronounce it differently,' meaning 'their pronunciations are incommensurable,' even if they sound so much alike that no difference can be heard.' This statement was met with amazed incredulity, and a request to state honestly whether anybody else held similar views. It is to be feared that the answer to this request was not accepted at face value. The trouble was, of course, that no bridge had been built between the engineer's knowledge of telegraphy and the language experience of linguists.

Now let us go on with the design of language. We have dealt with the phonemes individually, and shown that the most important thing about them is that they are either absolutely the same or absolutely different. And yet, in another sense it is to be expected that two of them can be partly the same and partly different. This can be possible in this way. If it is possible to put phoneme *A* into three categories *x*, *y*, and *z* (as the word *noes* simultaneously belongs to the three categories "noun," "common rather than possessive," and "plural"), and to put phoneme *B* into three categories *v*, *w*, and *z* (as the word *were* belongs to the three categories "verb," "past," and "plural"), then *A* and *B* can be said to be similar, but of course only in the sense that they are partly identical, the degree of similarity being quantized as demonstrated. These quantized partial identities will serve to build up the design of the phonemic system. Only one warning needs to be given: the worker should at each step ask himself whether it would make any difference whether he uses "component parts" or "intersecting categories" in dealing with such similarities. The phonetician is all too likely to use component parts in a field where intersecting categories are alone in order; that is one reason why the phonemic pattern was here presented parallel to the grammatical instance, for there is little temptation to speak of "plural" as component parts of the two words *noes* and *were*.

The Morse code uses two signalling units, "mark" and "space," differing in one dimension; or it can be considered as using four units, "dot," "dash," "space," and "letter-space," different in two dimensions, namely electrical intensity (or polarity) and duration. On the other hand, every known language uses more than a dozen or twenty (and less than a hundred) phonemes, differing from each other in at least four, or more probably at least five, dimensions. Spanish, for example, seems to have at least nine dimensions in its phonemic system, and English is known to have more than Spanish.

These are so many that a complete catalog would be boring and confusing; but a partial listing will help

make it clear what this is all about. One of the Spanish dimensions is "vowel-versus-consonant." Among the five Spanish vowels, two more dimensions are clearly evident. In one dimension, the five vowels are categorized as high, mid, and low; in the other, as front, central, and back. /i/ and /u/ are high, /e/ and /o/ are mid, and /a/ is low; /i/ and /e/ are front, /u/ and /o/ are back, and /a/ is neither, so it is called central. The six category-names were borrowed from phonetics into phonemics, just as the word "work" was borrowed by physicists, and as we borrowed "plural" from every-day terminology, where it was a synonym for "numerous." In phonetics they referred to tongue-position in the mouth; in phonemics, of course, we must be careful to strip away those denotations of the words, which have now become distracting connotations. In phonetics it is clear that *e* is intermediate between *i* and *a*. But in the phonemic structure of Spanish /e/ may or may not be intermediate between /i/ and /a/ on the evidence already presented here, for the articulatory facts are not what we need for deciding the question. In Finnish, for example, *y* is of course phonetically between *i* and *u*, and *ö* is between *e* and *o*; but in the phonemic system of Finnish there is compelling evidence (the diphthongs, for instance) to show that /i/ is between /y/ and /u/, and /e/ is between /ö/ and /o/.

Again telegraphy furnishes an analogy to show how this works. In submarine cable use of the Morse code, there are three signalling units, all of the same duration, ranged in the single dimension of electrical polarity: the land-line "dot" becomes a positive pulse, the land-line "dash" becomes a negative pulse, and the land-line "letter-space" becomes an electrical neutral interval, while the land-line short space drops out of the code. In practise, the "letter-space" unit is repeated for word-space, and repeated indefinitely (nothing is sent) for silence, representing for example suspension of transmission.

Now suppose we should rearrange the cable code again, by a transposition, so that "dot" is a neutral interval, "letter-space" is a positive pulse, and "dash" is still a negative pulse. At first sight it seems that this new code ought to work equally well, for it appears to be mathematically equivalent to the customary cable code. But it won't work at all, for the reason that an idle period, a pause between messages or the like, would load up the line with an unbroken sequence of positive pulses; the line would become heavily charged, which would block its functioning for a while when transmission was resumed again.

The same sort of thing is found in language, though of course we cannot explain "why." Spanish, for example, has as part of its design the fact that unaccented /i/ and /u/ do not occur after a consonant before pause (except in the queer word *espíritu* and perhaps a few other rare words). This fact and a few others of like sort will suffice to place /e/ between /i/ and /a/ in the phonemic system of Spanish. And this is what

usually turns up; patterns like the Finnish one are fairly rare.

A telegraphic "mark" does not signal anything by itself. In principle, it takes an assemblage of marks and spaces to make up a signal. Then the one-dot Morse *E* is a sort of accident, what the mathematicians call a "trivial" or "degenerate" case: it is a monatomic molecule, like a molecule of helium.

The same is true of phonemes. Linguistic signalling is done, not with phonemes, but with morphemes, like either morpheme of the word *noses*. It is only when the coding has been built up at least to the morpheme level that it is permissible to talk of "meaning." The morphemes, viewed from the phonemic level, are assemblies; viewed from levels of higher complexity, the morphemes appear as unanalyzable units; viewed from the outside, the morphemes are sub-assemblies in engineering terminology. They combine into larger sub-assemblies, and these again and again into more and more massive ones, until a complete utterance has been built up, through a number of stages of assembly. In the English sentence about the councillors, at least seven stages of assembly can be found, at least seven layers of complexity in the coding of a fairly simple message. We shall discuss them only summarily.

If we start with the complete utterance and break it down by stages until we arrive at the phonemes, a pattern of analysis emerges which has interesting peculiarities. As a whole it belongs to the category "sentence"; nothing less than all of it is thus classified; this classification is plainly marked by details of rhythm and intonation which the spelling does not indicate but which are essential structural details of English. Parts of it belong to different classes; larger parts of it may or may not belong to categories which smaller sub-assemblies cannot be put into. First analysis: cut it into the *councillors* and *all put their glasses on their noses*. The first part goes into several categories; similarly, in the four-unit analysis of Morse, the "dash" belongs to the two categories "long" and "mark." It is a "noun-phrase" because it fits into the sentence the same way as a plain "noun" might, such as the word *councillors*; its further analysis will of course agree with this, for example it contains a "noun" and the word *the*, and such combinations generally make up noun-phrases. It is an "actor-expression," so marked by the absence of any other candidate for the job, while the rest of the sentence, the "action-expression," can use one and will cause anything to be so classified which is not marked as having a different value: this one is rather marked as probably being an "actor-expression" by the fact that it is a "noun-phrase" and stands first, and its classification as an "actor-expression" is not forbidden by any such mark as the *s* of *man's*. Hence it is "common" rather than "possessive," although it is not marked as such, as the words *man* and *men* are: the apostrophe of *councillor's* or *councillors'* belongs to spelling, not to the language. Therefore the final *s* here

marks it as belonging to the "plural" category; this is at least not forbidden by the form *put*, while *puts* would refuse to fit it; therefore, when we later analyze the "action-expression" we shall classify *put* as "plural," whereupon it will be seen that *put* is not marked as either "present" or "past," while *puts* would have been marked as "present." It is marked as "definite" by the presence of *the* in it; being "definite," it will fit a "plural" "action-expression" containing the word *all* used as it is here, which at least some "indefinite" "actor-expressions" will not fit, for instance *some councillors*. In summary, this *the councillors* is a "common definite plural noun-phrase" functioning here as an "actor-expression"; it will not do simply to call it a "common definite plural noun-phrase actor-expression" because then "common" would be rendered redundant by "actor-expression." On the other hand, "actor-expression" is not made redundant by "common," for "common rather than possessive" will fit other categories besides "actor-expression," as in *I met the councillors*. It is precisely such asymmetries, found everywhere in language, which makes our analysis possible and keeps our arguments from being circular. The reader may check through this paragraph for other asymmetries and see how they were taken advantage of.

Second analysis: cut *the councillors* into *the* and *councillors*. The reader can set up the discussion himself on the model of the preceding paragraph.

Third analysis: cut *councillors* into *councillor* and *s*. Now *councillor* looks like a word, but not in this sentence; we can call it a "stem" here; to it is added the "morpheme" *s*. Discussion of this would use up our space unprofitably.

Fourth analysis: cut *councillor* into *council* and *or*. Either one of these is a morpheme. The first of them looks like the one-morpheme word *council* (compare the one-dot Morse *E*), but here it isn't; call it a "morpheme" and stop there. It belongs to the category of morphemes defined by the fact that they will combine with *or*, and possibly to other categories similarly defined by possibilities of combination, for example it will combine with *s* to make either the possessive singular *council's* or the plural *councils* or *councils'*. It will be noted that these criteria of classification are essentially of the same sort as those used in our first analysis, but now the categories are different ones.

Fifth analysis: we may or may not choose to break *council* into two syllables, but it appears that we can never get any profit out of doing so in English, so we cut it at once into seven phonemes—by an odd accident, the same as the number of letters in the spelling—by a single act of analysis.

Now we have arrived at the phoneme level, and no further analysis is called for by the kind of procedure we have been following. We can classify the phonemes into different categories of vowels and consonants, as was done above for Spanish. Then we could, if we chose, treat these intersecting categories as component

parts, and this work, known as "componential analysis," is done by some linguists. But although possible, this procedure does not often shed any light on the higher organization of the language, and for this reason even the practitioners of componential analysis normally use phonemic writing, not componential transcription, in preference to the orthographic style in which we cited the sentence about the councillors. Hence we may as well stop here and say that we have reached the lowest level of analysis when we have split *council* into seven phonemes. Here it was called the fifth analysis; to get to the bottom of the second half of the same sentence would have required going to a seventh analysis. This is the reason why it would be better to start numbering from the bottom, and to speak rather of successive syntheses, from phonemes up to the complete utterance, reaching different heights in different parts of it.

Now we note that each act of analysis except the bottom one was a dichotomy, a cutting into just two parts, never into three or more by a single step. And this has been found to be the best style of analysis in all languages, from the complete utterance down to the morpheme level; but in breaking morphemes up into phonemes, any number of components is possible from one on up. This looks like it might be connected with the point that "meaning" has nothing to do with phonemes, but belongs to the morphemics and higher levels; and this is conceivably connected with the functioning of the brain as a binary computer, but we daren't go into such matters.

Within an English morpheme some sequences of phonemes are forbidden. For example, what is commonly spelled *ou* (as in *council*) is never followed by *p*, *b*, *f*, *v*, or *m*. We have already pointed out, with illustrations from Spanish and Finnish, how such restrictions enable us to assign a shape to the phonemic system and define its categories. This one, for instance, defines the "labial" class of English consonants. We have shown further that analogous restrictions apply at every level of analysis up to the complete utterance. Above the level of the phonemic system, these restrictions define for us the "grammar" of the language, and all without reference to what is popularly thought of as the "meaning," namely the popular categorizations of continuous reality such as "nose" or "numerous," just as we find it expedient to define the phonemic system independently of the phonetician's categorizations of continuous speech.

Could we perhaps do something further with "meaning" inside linguistics? Yes, but only on condition that

we distinguish sharply between the inside and the outside. Let the sociologists keep the outside or practical meaning; then we can undertake to describe the pure linguistic meaning. We can do it thus:

Among permissible combinations of morphemes, some are commoner than others. Thus there are conditional probabilities of occurrence of each morpheme in context with others. The conditional probability of a forbidden occurrence is of course zero and drops out of the picture. If we ever found a conditional probability of one, we should decide that we had made a miscalculation, and that what we thought was composed of two morphemes was really unanalyzably one morpheme.

Now the linguist's "meaning" of a morpheme (or of a combination thereof, of any complexity up to a complete utterance, which might even be a whole book or poem) is by definition the set of conditional probabilities of its occurrence in context with all other morphemes—of course without inquiry into the outside, practical, or sociologist's meaning of any of them. Of course all zero probabilities drop out of the set automatically; and in practise we should neglect all very small probabilities. So far we have done almost nothing with pure linguistic "meaning" as so defined, for the obvious reason that its mathematics is of the continuous sort, which we are not accustomed to handling—continuous by statistical derivation from discreteness, but still unmanageable, so far, under our working-habits. Still, a beginning has been made on a structural semantics by one linguist. He replaces all small conditional probabilities by zero, and all large ones by one, so as to get a discrete mathematics again, in which he can work with synonymy that is absolute by definition. This work is very recent, and is the most exciting thing that has happened in linguistics for quite a few years; in spite of its theoretical flaw, it has produced some very elegant and illuminating results.

We can close with a general characterization of linguistics and language. Most well-known mathematical maps are connected with the real world by a single intuitive bridge; linguistics is connected with reality by two of them, so that language makes a real noise correspond to a real thing or the like. This defines language as symbolic. Overt acts of speech alter reality in that changes in listener's behavior correspond to them. This defines speech as communication. The technique is organized, patterned, as disclosed by linguistic science. This defines language as systematic. Altogether, then, we have defined a language as a symbolic communication system, or in one word, as a "code."